Exact Renewal and Clock Recovery for Odd-Affine Parity Maps on the 2-Adic Integers

Route-A structural certificate HCS-C174

Abstract

For odd integers a, b, consider the parity map on the 2-adic integers that sends an even x to $x/2$ and an odd x to $(ax+b)/2$. Classical parity coding conjugates this family to the one-sided binary shift; we use that conjugacy as prior foundation, not as a novelty claim. We derive an exact first-return renewal description on the odd coset. The return time is $\tau(x) = v_2(ax+b)$ with conditional Haar law 2^{-k}, and the return map is a countable full shift with roof $r(k) = k$. Its ordinary Artin–Mazur series is unavailable because it has infinitely many fixed points, but the roof recovers the original clock: $(1 - \sum_{k\geq 1} z^k)^{-1} = (1-z)/(1-2z)$, and restoring the zero orbit gives $(1-2z)^{-1}$. Moreover both the unweighted fixed counts and reciprocal 2-adic derivative stability sums are independent of every odd parameter pair. Exact solvability therefore supplies a parameter-blindness obstruction, not an arithmetic target model.

Keywords: 2-adic dynamics; parity coding; first return; renewal shift; dynamical zeta; Koopman isometry.

中文摘要

对任意奇整数 a, b，本文研究二进整数环上的奇偶分支映射：偶点映为 $x/2$，奇点映为 $(ax+b)/2$。经典 parity 编码把该系统共轭到单边二元 移位；本文把这一事实明确视为先验基础，而非新颖性主张。我们在奇数陪集上 推导精确首返 renewal：返回时间为 $\tau(x) = v_2(ax+b)$，条件 Haar 分布为 2^{-k}，首返映射是带屋顶 $r(k) = k$ 的可数字母满移位。加速映射因时间一 已有无限多个不动点而不存在普通 Artin–Mazur 级数；但屋顶精确恢复原始时钟， 再补回零轨道即得 $(1-2z)^{-1}$。同时，无权不动点计数与二进导数稳定性权重 对所有奇参数都完全失明。因此，精确可解性给出的是参数盲性障碍，而非算术目标模型。

1 Frozen family and ownership boundary

Fix odd $a, b \in \mathbb{Z}$ and normalized Haar probability μ on $\mathbb{Z}_2$. The one-tick map is

$$T_{a,b}(x) = \begin{cases} x/2, & x \equiv 0 \pmod 2, \\ (ax+b)/2, & x \equiv 1 \pmod 2. \end{cases} \tag{1}$$

Oddness is structural: each numerator in the active branch is even. The parity vector $Q_{a,b}(x) = \sum_{j\geq 0} \epsilon_j(x) 2^j$, with $\epsilon_j(x) = T^j(x) \bmod 2$, conjugates $T_{a,b}$ to the dyadic shift. Bernstein and Lagarias treated this conjugacy and its odd $ax+b$ generality [1]. We claim no priority for it. The question here is what first-return acceleration does to clock and parameter information.

If $s_j = \sum_{i<j} \epsilon_i$, the inverse parity series is

$$Q_{a,b}^{-1}\left(\sum_{j\geq 0} \epsilon_j 2^j\right) = -b \sum_{j\geq 0} \epsilon_j 2^j a^{-s_{j+1}}. \tag{2}$$

It also shows that $Q_{a,b}$ preserves congruence cylinders and hence Haar measure. Multiplication by b conjugates $T_{a,1}$ to $T_{a,b}$.

2 Fixed words and stability blindness

For a word $\epsilon = (\epsilon_0, \ldots, \epsilon_{n-1})$, put

$$A_\epsilon = \sum_{j=0}^{n-1} \epsilon_j 2^j a^{s_n - s_{j+1}}.$$

Branch induction gives

$$2^n T^n(x) = a^{s_n} x + b A_\epsilon. \tag{3}$$

Since $2^n - a^{s_n}$ is odd, every word has the unique fixed point

$$x_\epsilon = \frac{b A_\epsilon}{2^n - a^{s_n}} \in \mathbb{Z}_2. \tag{4}$$

The parity conjugacy makes all 2^n words admissible and distinct. Thus

$$\#\operatorname{Fix}(T^n) = 2^n, \quad P(n) = \sum_{d|n} \mu(n/d) 2^d, \quad \zeta_{\mathrm{AM}}(z) = \frac{1}{1-2z}. \tag{5}$$

Here $P(n)$ counts exact-period points, so $P(n)/n$ counts primitive cycles.

The branch derivatives are $1/2$ and $a/2$. On the point with word ϵ,

$$(T^n)' = \frac{a^{s_n}}{2^n}, \qquad \left|1 - \frac{a^{s_n}}{2^n}\right|_2 = 2^n,$$

because $2^n - a^{s_n}$ is odd. Consequently

$$W_n := \sum_{x \in \operatorname{Fix}(T^n)} |1 - (T^n)'(x)|_2^{-1} = 1, \qquad \zeta_{\mathrm{stab}}(z) = \frac{1}{1-z}. \tag{6}$$

Equations (5) and (6) are all-parameter theorems: both invariants erase a and b.

3 First-return renewal and clock recovery

Let $O = 1 + 2\mathbb{Z}_2$. For $x \in O$ with $ax + b \neq 0$, define

$$\tau(x) = v_2(ax+b), \qquad R(x) = \frac{ax+b}{2^{\tau(x)}}. \tag{7}$$

The point $x_* = -b/a$ maps to zero and has infinite return time. More generally, let E be the set of odd points whose parity strings contain only finitely many ones. Under the parity homeomorphism, E is the union over all finite binary prefixes of one eventually-zero tail; it is therefore countable and Haar-null, and it contains x_*. Its complement $O_\infty = O \setminus E$ is invariant under successive returns and has full conditional Haar measure.

The affine map $x \mapsto ax + b$ sends O bijectively and measure-preservingly onto $2\mathbb{Z}_2$. Hence the normalized Haar layers give

$$\mu_O(\tau = k) = 2^{-k}, \qquad k \geq 1. \tag{8}$$

The parity block associated with k is 10^{k-1}. Concatenation gives a bijection from O_∞ to $\{1, 2, \ldots\}^{\mathbb{N}}$, conjugating R to the left shift. Bernoulli independence of disjoint parity blocks then makes successive returns iid with the geometric product measure. The constant block k has fixed point

$$x_k = \frac{b}{2^k - a}. \tag{9}$$

Thus R already has infinitely many time-one fixed points and its ordinary Artin–Mazur series is undefined.

Return symbol k represents k original ticks. With roof $r(k) = k$, the first-return series and renewal zeta are

$$F(z) = \sum_{k \geq 1} z^k = \frac{z}{1-z}, \qquad \zeta_{\mathrm{roof}}(z) = \frac{1}{1 - F(z)} = \frac{1-z}{1-2z}. \tag{10}$$

Indeed $\log \zeta_{\mathrm{roof}} = \sum_{n \geq 1} (2^n - 1) z^n / n$: these are exactly the original-clock periodic points whose binary words contain a one. The omitted all-zero fixed orbit contributes $(1-z)^{-1}$, and therefore

$$\zeta_{\mathrm{roof}}(z)(1-z)^{-1} = \frac{1}{1-2z} = \zeta_{\mathrm{AM}}(z). \tag{11}$$

4 Operator boundary and Route-A decision

On $H = L^2(\mathbb{Z}_2, \mu)$, $Uf = f \circ T$ is an isometry. It is not surjective: under parity coding, its range consists of functions independent of the first digit. Thus it is neither compact nor trace class, and no ordinary Fredholm determinant $\det(I - zU)$ is available. Koopman methods have also been studied for the discrete positive-integer $3x+1$ system [2]; we make no general priority claim for a Collatz Koopman lift.

The exact Route-A tuple is

(`A0_FAIL`, `A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_FORMAL_HINT`).

A0 fails because dyadic local arithmetic supplies no rational-prime or prime-power correspondence. A1 is weak despite the complete primitive ledger because the orbits carry no arithmetic labels. The elementary, parameter-blind zetas fail A2 and A3. The non-trace-class isometry is only an A4 formal hint. A0 failure forces overall rejection.

5 Limitations and reproducibility

For $(a, b) = (3, 1)$ the 2-adic word 100 gives $1/5 \mapsto 4/5 \mapsto 2/5 \mapsto 1/5$. This legal $\mathbb{Z}_2$ cycle is not a positive-integer orbit, so the paper gives no progress on the positive-integer $3x + 1$ conjecture. If exactly one of a, b is even, the odd-branch numerator is odd and division by two leaves $\mathbb{Z}_2$; the theorem does not cross that boundary.

The release uses exact arithmetic only. A producer, independent checker, SymPy derivation, byte replay, semantic mutation suite, and content-addressed manifest are included. Finite ledgers are regression sentinels, not proofs. No prime table, target zero or divisor, arithmetic local factor, Euler factor, root number, automorphy, Hilbert–Pólya operator, Route-B authorization, external review, or acceptance rate is claimed. The scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`.

中文边界说明。 本文的精确结论限定于全体奇参数的 $\mathbb{Z}_2$ 动力学。经典 parity 共轭属于先验工作；有限回归表不替代理论证明；$(3, 1)$ 特例不推出正整数 Collatz 结论。由于 A0 缺少素数及素数幂来源，整体 Route A 必须拒绝。

References

[1] D. J. Bernstein and J. C. Lagarias, "The 3x+1 Conjugacy Map," *Canadian Journal of Mathematics* 48 (1996), 1154–1169. doi:10.4153/CJM-1996-060-x.

[2] J. Leventides and C. Poulios, "Koopman operators and the $3x + 1$-dynamical system," arXiv:2010.12987 (2020). arxiv.org/abs/2010.12987.